MODULE 1: PRINCIPLES OF ELECTRICAL ENGINEERING: CIRCUITS & POWER CONVERSION EQUIPMENT

(CO: 1)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with Official VIT BACEE101 Course Page Syllabus, Question Bank, and Past FAT Papers

Scope of This Textbook (Official VIT Syllabus - All 13 Topics): (1) Course Introduction & Review of circuit basics, (2) Mesh Current Analysis, (3) Nodal Voltage Analysis, (4) Thevenin's theorem, (5) Superposition theorem, (6) Single-phase RL, RC and RLC series circuits, (7) Power and Energy calculations, Power factor, (8) Basics of Electrical Safety and Earthing, (9) Principle of DC Motor, (10) Principle of 3-phase Induction Motor, (11) Principle of BLDC Motor, (12) Principle of Single-phase Transformer, and (13) Power electronics concepts: DC-DC Buck converter.

1 VIT FAT Exam Blueprint & Master Formula Sheet

Module 1 carries **20 to 25 marks** in BACEE101 FAT:

1. **Question Type A (10 Marks):** Circuit Analysis: Finding branch currents/voltages using Mesh/Nodal analysis or Thevenin's Theorem with resistive loads.
2. **Question Type B (10 Marks):** AC Series RLC Circuit: Impedance triangle, phasor diagram, active (P), reactive (Q), apparent (S) power, and power factor correction.
3. **Question Type C (10 Marks):** Electrical Machines: Operating principle, EMF/Torque equations of DC Motor, 3-Phase Induction Motor (slip calculations), Single-phase Transformer, or DC-DC Buck converter duty cycle.

Topic / Concept	Governing Equation / Formula	VIT Exam Key Insight
Thevenin Equivalent	$I_L = \frac{V_{\text{th}}}{R_{\text{th}}+R_L}$	V_{th} = open circuit voltage; R_{th} found by deactivating independent sources.
AC Series RLC Impedance	$Z = \sqrt{R^2 + (X_L - X_C)^2}$, $\tan\varphi = \frac{X_L - X_C}{R}$	$X_L = 2\pi f L$, $X_C = \frac{1}{2\pi f C}$; resonance when $X_L = X_C$.
AC Power Relations	$P = VI\cos\varphi$, $Q = VI\sin\varphi$, $S = VI = \sqrt{P^2+Q^2}$	Power factor is $\cos\varphi = \frac{P}{S}$; lagging for inductive, leading for capacitive.
DC Motor Back-EMF	$E_b = \frac{P\Phi ZN}{60A} = V - I_a R_a$	Torque $T \propto \Phi I_a$; speed $N \propto \frac{E_b}{\Phi}$.
Induction Motor Slip	$N_s = \frac{120f}{P}$, $s = \frac{N_s - N}{N_s}$, $f_r = sf$	Rotor speed N is always strictly less than synchronous speed N_s.
Transformer EMF Equation	$E_1 = 4.44 f N_1 \Phi_m$, $E_2 = 4.44 f N_2 \Phi_m$	Transformation ratio $K = \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{I_1}{I_2}$.
DC-DC Buck Converter	$V_{\text{out}} = DV_{\text{in}}$, $D = \frac{T_{\text{on}}}{T} = \frac{T_{\text{on}}}{T_{\text{on}}+T_{\text{off}}}$	Step-down DC voltage; duty cycle D satisfies $0 \le D \le 1$.

2 Circuit Analysis & Network Theorems

2.1 Mesh Current & Node Voltage Analysis

- **Kirchhoff's Current Law (KCL):** Algebraic sum of currents entering any electrical node is zero: $\sum I = 0$.
- **Kirchhoff's Voltage Law (KVL):** Algebraic sum of all potential differences around any closed circuit mesh is zero: $\sum V = 0$.

- **Thevenin's Theorem:** Any linear bilateral two-terminal network consisting of voltage sources, current sources, and resistors can be replaced by an equivalent circuit containing an ideal voltage source V_{th} in series with a resistance R_{th}.

2.2 Single-Phase AC Series Circuits

For a series connection of resistor R, inductor L, and capacitor C energized by an AC sinusoidal voltage $v(t) = V_m \sin(\omega t)$:

- Inductive reactance: $X_L = \omega L = 2\pi f L$.
- Capacitive reactance: $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$.
- Net impedance: $\boldsymbol{Z} = R + j(X_L - X_C) = |Z| \measuredangle \varphi$, where $|Z| = \sqrt{R^2 + (X_L - X_C)^2}$ and phase angle $\varphi = \arctan\left(\frac{X_L - X_C}{R}\right)$.
- **Real Power (P):** $P = V_{\text{rms}} I_{\text{rms}} \cos\varphi$ (measured in Watts, W).
- **Reactive Power (Q):** $Q = V_{\text{rms}} I_{\text{rms}} \sin\varphi$ (measured in Volt-Amperes Reactive, VAR).
- **Apparent Power (S):** $S = V_{\text{rms}} I_{\text{rms}}$ (measured in Volt-Amperes, VA).

3 Electrical Machines & Power Electronics

3.1 Single-Phase Transformer

A static electrical device that transfers electrical power between two circuits via mutual electromagnetic induction without changing frequency.

- **Faraday's Law of Induction:** Induced EMF $e = -N\left(d\frac{\Phi}{d}t\right)$.
- Assuming sinusoidal flux $\Phi = \Phi_m \sin(\omega t)$:

$$e = -N\left(\frac{d}{dt}[\Phi_m \sin(\omega t)]\right) = -N\omega\Phi_m \cos(\omega t) = N(2\pi f)\Phi_m \sin\left(\omega t - \frac{\pi}{2}\right)$$

RMS induced EMF:

$$E = \frac{2\pi f N \Phi_m}{\sqrt{2}} = 4.44 f N \Phi_m$$

$$E_1 = 4.44 f N_1 \Phi_m \quad \text{and} \quad E_2 = 4.44 f N_2 \Phi_m$$

3.2 DC Motors & BLDC Motors

- **DC Motor Principle:** Current-carrying conductor in a magnetic field experiences a Lorentz force $F = BIl$ (Fleming's Left-Hand Rule). Rotation induces back EMF $E_b = \frac{P\Phi ZN}{60A}$. Terminal voltage: $V = E_b + I_a R_a$.
- **BLDC (Brushless DC) Motor:** Replaces mechanical brushes and commutator segments with electronic switching (inverter bridge) and permanent magnet rotor. Hall effect sensors or back-EMF sensing monitor rotor position to trigger sequential stator windings, delivering high efficiency, low maintenance, and long lifespan.

3.3 DC-DC Buck Converter

A switch-mode power supply topology that steps down an unregulated DC input voltage V_{in} to a lower regulated DC output voltage V_{out}:

- **Switch ON ($0 < t < DT$):** Diode is reverse-biased; inductor current ramps up, storing magnetic energy: $v_L = V_{\text{in}} - V_{\text{out}}$.
- **Switch OFF ($DT < t < T$):** Diode freewheels; inductor discharges energy into load: $v_L = -V_{\text{out}}$.
- By volt-second balance across the inductor in steady-state:

$$(V_{\text{in}} - V_{\text{out}})DT + (-V_{\text{out}})(1 - D)T = 0 \Rightarrow V_{\text{out}} = DV_{\text{in}}$$